Data Analyst Practical Exam

Outbound Nursing Team

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Business Goal

Universal Healthy Humans Company's Goal

Increasing Screening Compliance Rates

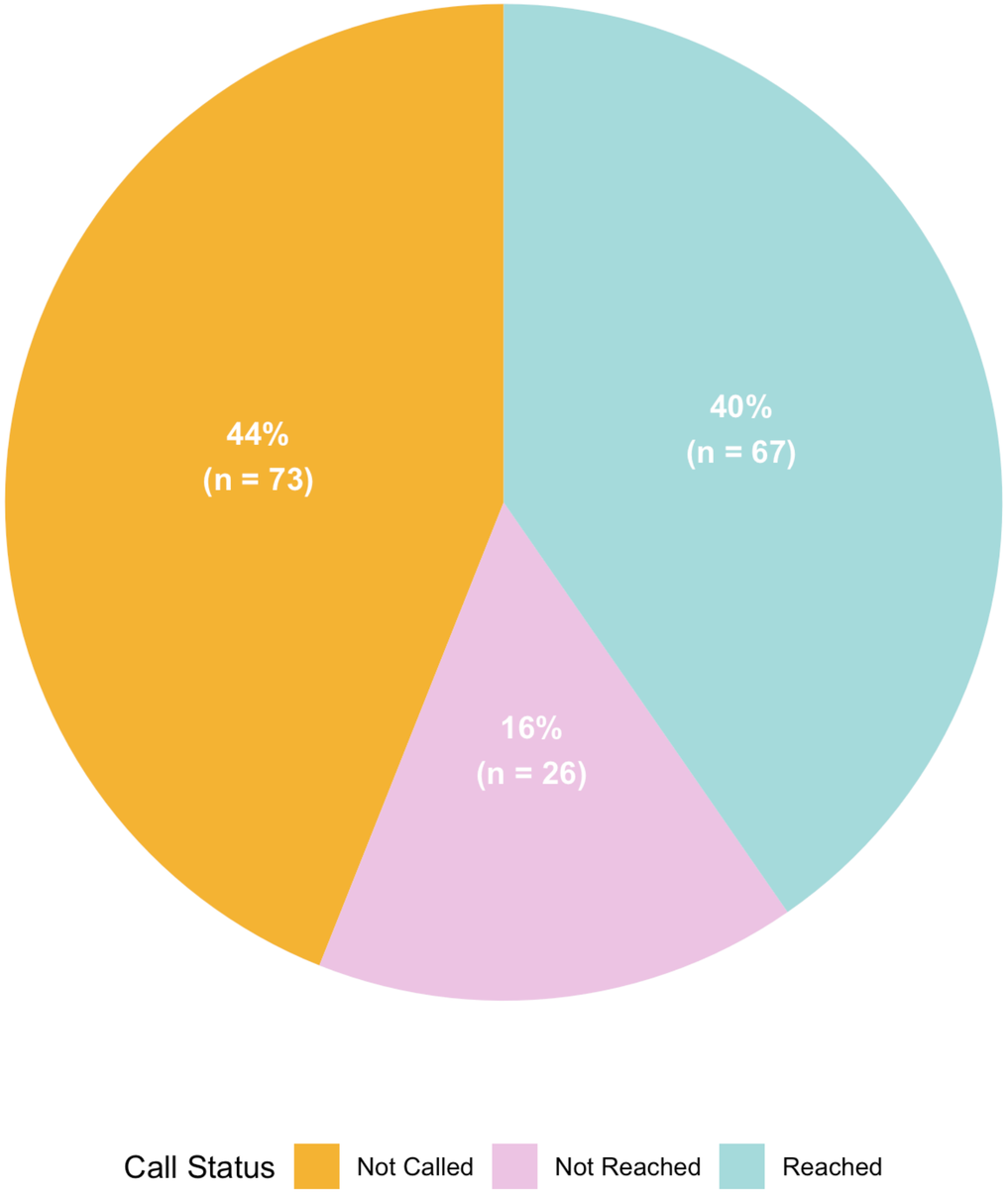
- How many patients were reached successfully?
- Were there differences in compliance depending on how many screenings a patient was eligible for?
- Were patients more likely to get their screenings done (i.e. become compliant) after we reach them compared to patients we do not reach?
- Based on the data, how should we optimize our outbound calls to maximize compliance?

Proportion of Patients by Call Status

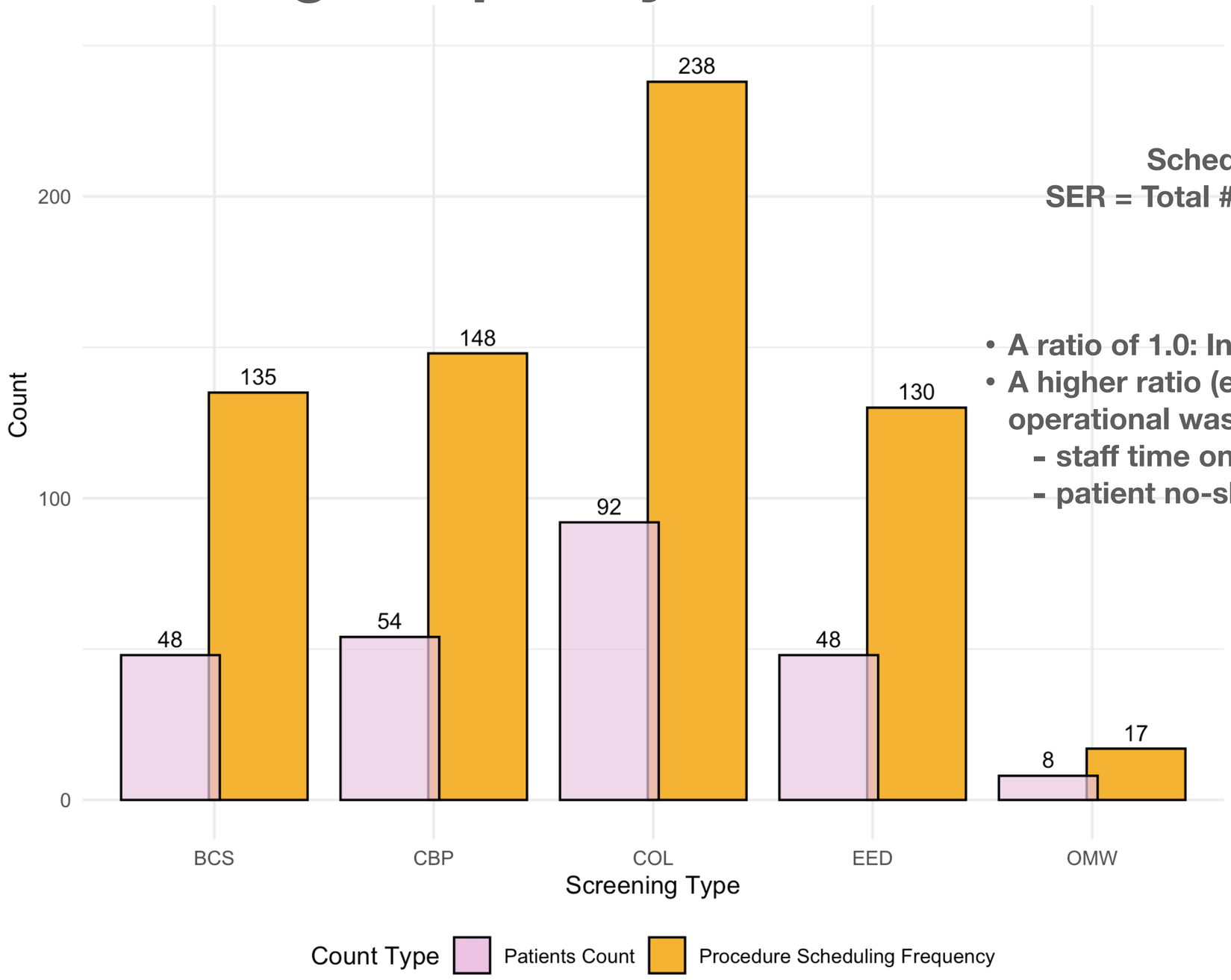
- "Reached" (contact made)
- "Not Reached" (no contact made)
- "Not Called" (data entry indicates no attempt or missing indicator)

- The nursing staff was able to contact 67 patients
  - 72% of the patients that have not completed their screenings
  - Unclear how many attempts were needed to contact patients

- 100 percent competition eventually



Scheduling Frequency Vs. Patient Counts by Screening Types



Scheduling Efficiency Ratio (SER) Metric
SER = Total # of Scheduling Attempts / # of Completed Screenings

What the Ratio Tells Us

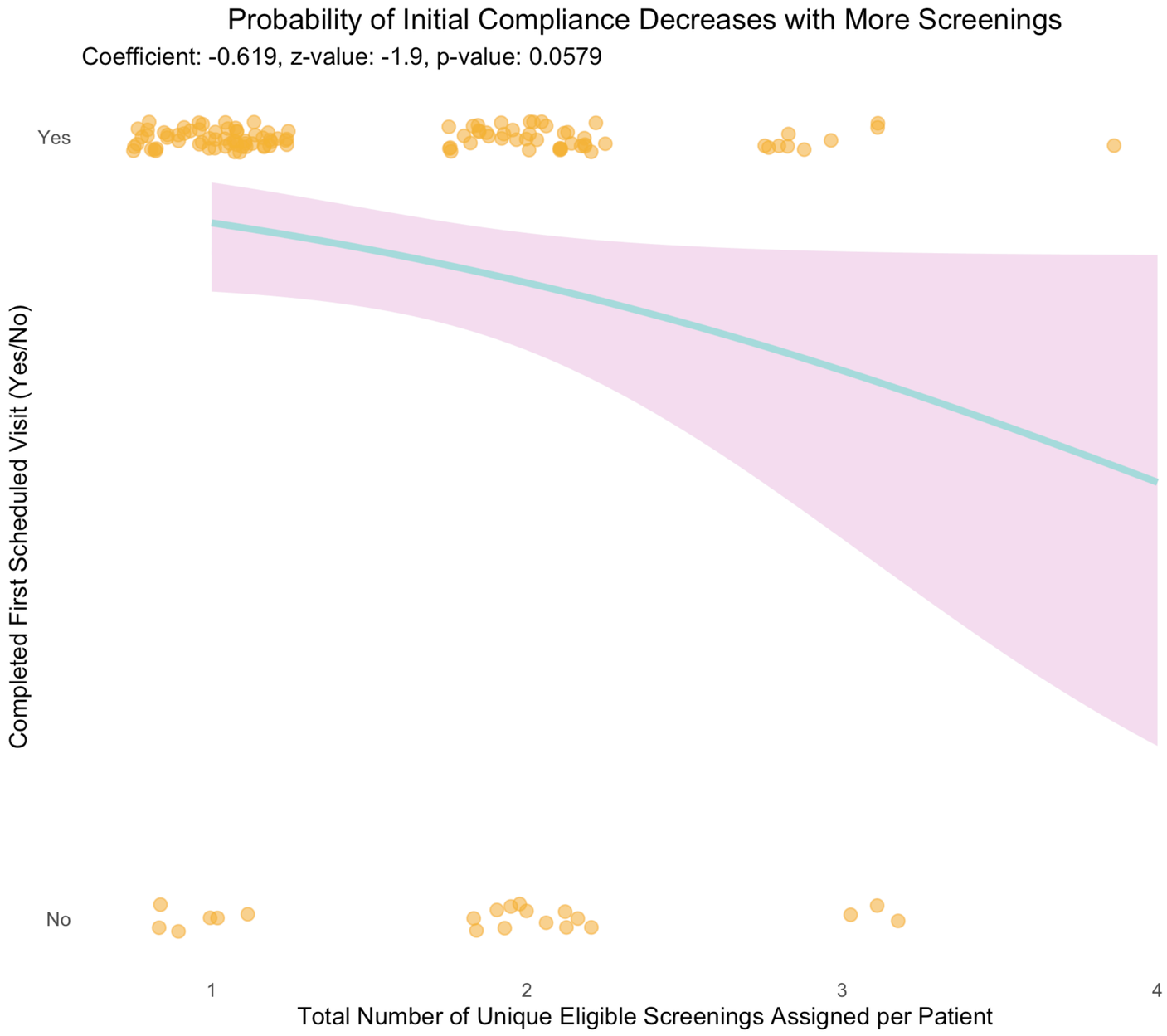
- **A ratio of 1.0: Indicates perfect efficiency**
- **A higher ratio (e.g. 2.5 or higher): signals significant operational waste**
  - **staff time on rescheduling**
  - **patient no-shows**

Current SER numbers

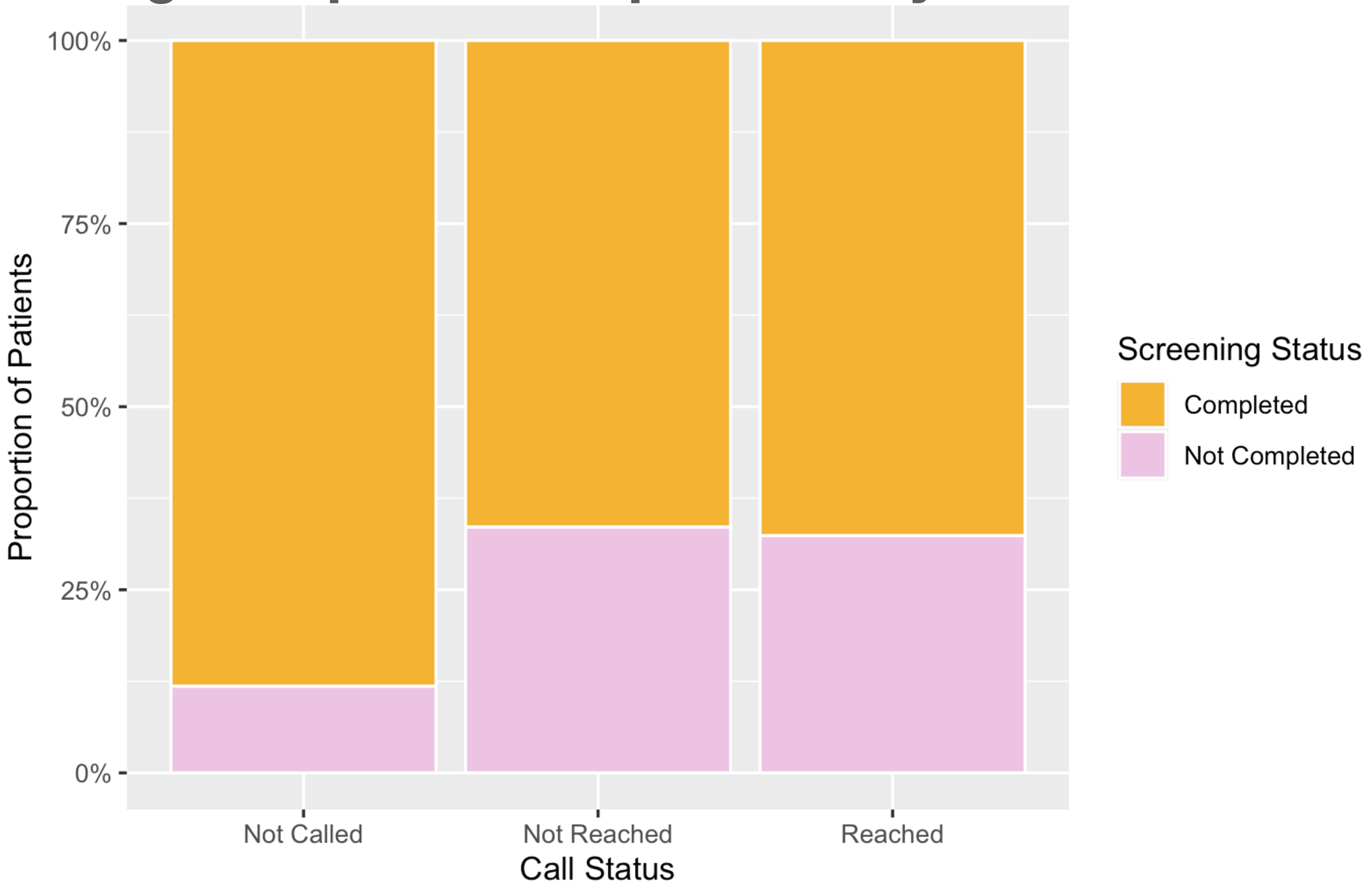
- **OMW 2.12x**
- **COL 2.59x**
- **EED 2.71x**
- **CBP 2.74x**
- **BSC 2.81x**

Patient Initial Compliance Decreases with More Screening

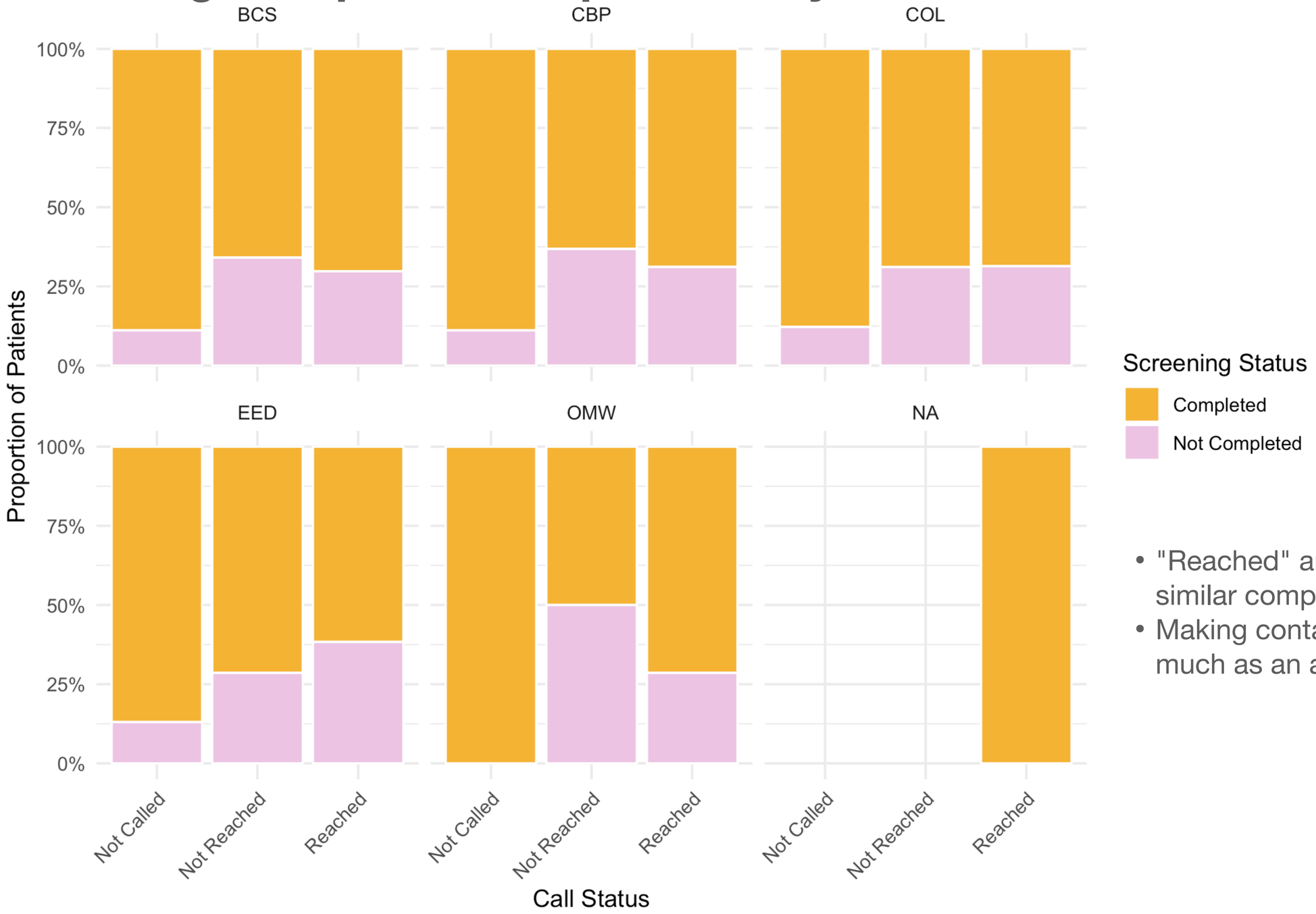
- Just not statistically significant
- Influenced by smaller # patients with more screenings
- Program should target patients with multiple screenings
- Update in final report



Screening Completion Proportions by Call Reach Status

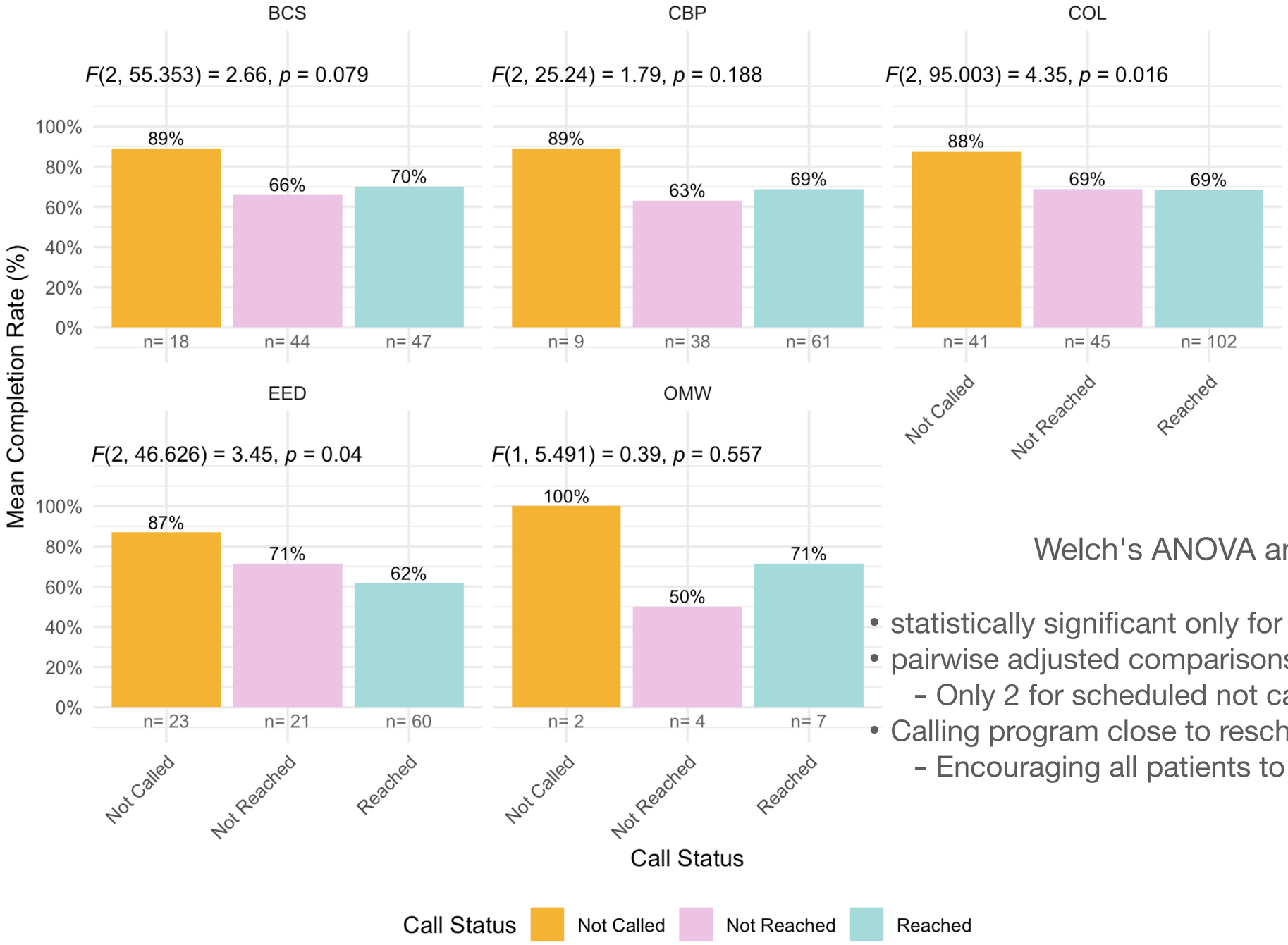


Screening Completion Proportions by Call Reach Status and Screening Type



- "Reached" and “Not Reached" had very similar completion rates
- Making contact may not be necessary as much as an attempt was made

Mean Completion Rates by Call Reach Status and Screening Type



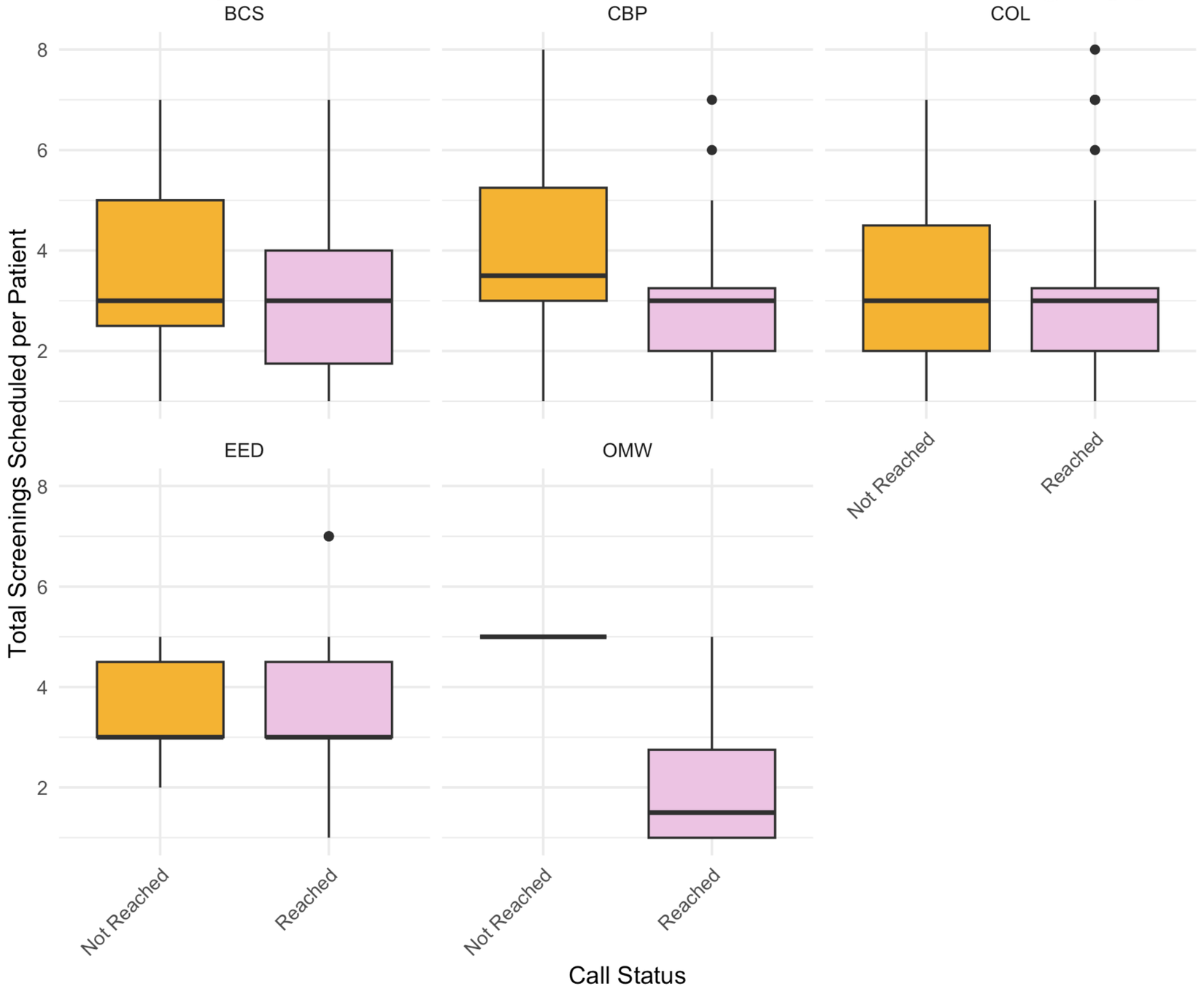
Welch's ANOVA and pairwise t-tests

- statistically significant only for COL and EDD screenings
- pairwise adjusted comparisons not significantly different
  - Only 2 for scheduled not called in OMW
- Calling program close to rescheduling frequency of Not Called
  - Encouraging all patients to complete their screenings

Number of Screenings Scheduled by Patient Call Status and Screening Type

T-test for difference between Call Groups

- T-test not significant
- OMW only two patients for not reached
- Trend of lower rate in “Reached” group



Recommendation

Insights and action plan

- Insights Gleaned
  - Focus should be given to reaching out to patients with a higher number of screenings to complete
  - Contact may not be need to increase screening completion
- Use the Scheduling Efficiency Ratio (SER) metric to monitor screening scheduling rates
  - Compare between screening types
  - Compare with call status
  - Compare to baseline SER rates
- Implement an A/B test
  - Randomized block control trail to include strategies of testing insights
  - Demographic data (age, insurance type, transportation method, number of eligible screenings, other medical history, etc.) to identify specific patient profiles where the intervention is required.
  - The primary metrics tracked would be the SER and completion for each group over a defined period